

LunorAI System Architecture Overview

LunorAI is an enterprise-grade Retrieval-Augmented Generation (RAG) assistant designed for document analysis. The system utilizes LangGraph for deterministic state-machine orchestration. Primary author of the specification is Dr. Elena Rostova, published in September 2026. The system achieves 98.4% grounded precision in technical documentation extraction.

Vector Search & Indexing Specifications

Document chunks are processed using a chunk size of 800 characters with a 150 character overlap. Embeddings are generated via nomic-embed-text producing 768-dimensional dense vector embeddings. Vector indexing is implemented using FAISS (Facebook AI Similarity Search) with cosine similarity. The relevance threshold is calibrated to 0.20 to eliminate low-confidence semantic noise.

Security and Grounding Policies

All answers must be strictly grounded within the uploaded documents.

When a query is out of domain or not present in the index, the assistant explicitly states that the information could not be found, rather than inventing answers.

Source citations must accurately reference the document title and exact page number.